Combinations

As introduced above, **combinations** are the different arrangements you can make from a set when order does not matter. Suppose you need to arrange the letters A, C, and B. The arrangements of ACB and ABC would be considered as one combination.

When dealing with more complex problems, we use the following formula to calculate combinations:

$$_nC_r = \frac{n!}{r!(n-r)!}$$

where n represents the total number of items, and r represents the number of items being chosen at a time.

Consider the following problem:

Katya has a jar with 5 different kinds of cookies. In how many ways can Katya choose 3 different cookies from the jar?

Answer:

We must first determine what type of question we are dealing with. Since order was not included as a restriction, we see that this is a combination question. Let us first determine our n and r:

$\implies n = 5$

there are 5 cookies

$\implies r = 3$

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we are choosing 3 cookies

$$\implies {}_nC_r = {}_5C_3 = 10$$

applying our formula

$\implies$ There are 10 ways in which Katya can choose 3 different cookies from the jar.

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Permutations

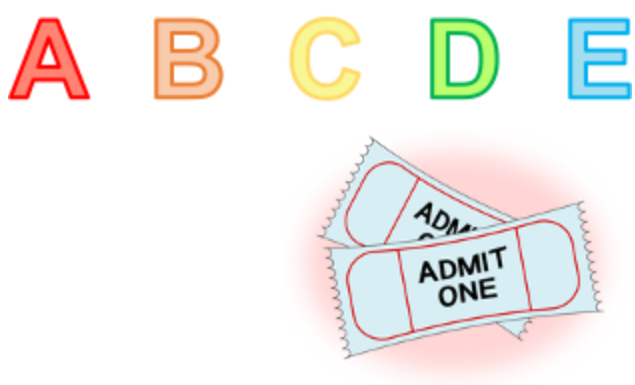
As mentioned in the introduction to this guide, **permutations** are the different arrangements you can make from a set when order matters. Suppose you need to arrange the letters A, C, and B. The arrangements of ACB and ABC would be considered as two different permutations.

When dealing with more complex problems, we use the following formula to calculate permutations:

$$_nP_r = \frac{n!}{(n-r)!}$$

Consider the following problem:

A football match ticket number begins with three letters. If the possible letters are A, B, C, D and E, how many different arrangements of these letters can be made if no letter is used more than once?



Answer:

We will solve this question in two separate ways.

<u>Method 1: Permutation Formula</u>

Let us first determine our n and r:

$\implies n = 5$

there are 5 letters

$\implies r = 3$

we are choosing 3 letters

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$\implies {}_nP_r = {}_5P_3 = 60$

applying our formula

$\implies$ There are 60 different arrangements of these letters that can be made.

Method 2: Logic

Let us break down the question into parts. For the first letter, we have 5 possible choices out of A, B, C, D, and E. After that letter is chosen, we now have 4 possibilities for the second letter. Finally, when choosing the third letter we are left with 3 possibilities. Using multiplication:

$$5 \times 4 \times 3 = 60$$

There are 60 different arrangements of these letters that can be made.

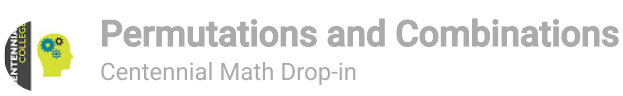
Try it yourself!

EXAMPLES

1. You just got a free ticket for a cinema hall, and you can only bring along 2 friends -but you have 5 friends who want to come with you. How many different groups of friends could you potentially take?

2. Meiqi has 6 different types of plants: cactus, rose, lily, cherry, poppy and jasmine. She has only 5 slots for planting in her garden. If these slots are in one single line, then in how many ways can she arrange her plants?

See the video below for the solutions:



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